

Canonicalization: Linearization Step

Original DCP

$$\begin{aligned} \min & f(\mathbf{x}) \\ \text{s.t. } & g_i(\mathbf{x}) \leq 0 \end{aligned}$$

Linearize
(introduce
atom vars)



Linearized (Canonical) Form

$$\min_{\mathbf{x}, \mathbf{v}} d + \sum_i (\mathbf{c}^{(i)})^\top \mathbf{x}^{(i)}$$

$$\text{s.t. } \mathbf{x}^{(i)} \in S^{(i)}$$

$$\sum_i \mathbf{A}^{(i)} \mathbf{x}^{(i)} = \mathbf{b}$$

$$f_j(\varepsilon_j) \leq v_j \text{ (convex atoms)}$$

$$f_j(\varepsilon_j) \geq v_j \text{ (concave atoms)}$$